

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0235

Rec Harmony 345 kV
March 03, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0235. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Rec Harmony 345 kV substation in the West Virginia area. The request was received on May 11, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 151 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$107,373,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Rec Harmony 345/138kV TX
Contingency: Rec Harmony to Dillard-Orange 345.0 kV Line
Loading: 149.5% (Rating: 449 MVA)
DFAX: 34.3%
Pre-project loading: 34.9%
2. Facility: Rec Harmony Substation to N Adams 138.0 kV line
Contingency: Dillard-Orange to Rec Harmony 345.0 kV line
Loading: 136.2% (Rating: 176 MVA)
DFAX: 16.0%
Pre-project loading: 0.2%
3. Facility: Huston to Grana 1907-1909 Substation 138.0 kV
Contingency: Dillard-Orange to Fishkill Plains 345.0 kV
Loading: 110.5% (Rating: 225 MVA)
DFAX: 7.0%
Pre-project loading: 63.8%
4. Facility: Dillard-Orange to Rec Harmony 345.0 kV
Contingency: Rec Harmony 345/138 kV Transformer Bank
Loading: 110.4% (Rating: 1,031 MVA)
DFAX: 69.8%
Pre-project loading: 8.8%
5. Facility: Grana 1907-1909 to Pana N 138.0 kV
Contingency: Fishkill Plains to Dillard-Orange 345.0 kV
Loading: 108.6% (Rating: 235 MVA)
DFAX: 7.0%
Pre-project loading: 64.0%

6. Facility: Pana N to Rec Harmony 138.0 kV
Contingency: Dillard-Orange to Fishkill Plains 345.0 kV
Loading: 106.8% (Rating: 246 MVA)
DFAX: 7.0%
Pre-project loading: 64.2%
7. Facility: Fishkill Plains to Dillard-Orange 345.0 kV
Contingency: Flatlick to International Paper 500.0 kV
Loading: 106.6% (Rating: 1,399 MVA)
DFAX: 34.0%
Pre-project loading: 70.1%
8. Facility: Medora to Hampton Rd 345.0 kV Line
Contingency: Fishkill Plains to Simpson Stuart Tap 345.0 kV L
Loading: 105.7% (Rating: 1,243 MVA)
DFAX: 11.4%
Pre-project loading: 92.0%
9. Facility: Medoa to P.T.S 345.0 kV
Contingency: Ritter Park to Sugar Hill 345.0 kV Line
Loading: 103.4% (Rating: 1,258 MVA)
DFAX: 12.5%
Pre-project loading: 88.5%
10. Facility: Sugar Hill to Ritter Park 345.0 kV
Contingency: Piney Crk Project to Hilltop Substation 500.0 kV
Loading: 101.5% (Rating: 1,250 MVA)
DFAX: 8.7%
Pre-project loading: 91.1%
11. Facility: Stewart Mtn Dam to Ritter Park 345.0 kV
Contingency: P.T.S to Medora 345.0 kV
Loading: 100.6% (Rating: 1,173 MVA)
DFAX: 10.4%
Pre-project loading: 87.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rec Harmony Substation 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$20,541,000	100% (\$20,541,000)
Rec Harmony Substation to N Adams 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,840,000	100% (\$4,840,000)
Huston to Grana 1907-1909 Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$14,357,000	100% (\$14,357,000)
Dillard-Orange to Rec Harmony 345.0 kV	Rebuild line with higher-capacity conductor	\$40,114,000	100% (\$40,114,000)
Grana 1907-1909 to Pana N 138.0 kV	Reconductor line	\$2,338,000	100% (\$2,338,000)
Rec Harmony to Pana North 138.0 kV line	Reconductor line	\$1,493,000	100% (\$1,493,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dillard-Orange to Fishkill Plains 345.0 kV	Reconductor line	\$33,452,000	47% (\$15,705,000)
Medora to Hampton Rd 345.0 kV Line	Reconductor line	\$13,406,000	20% (\$2,623,000)
Medoa to P.T.S 345.0 kV	Reconductor line	\$10,283,000	29% (\$3,020,000)
Sugar Hill to Ritter Park 345.0 kV	Reconductor line	\$4,685,000	22% (\$1,054,000)
Stewart Mtn Dam to Ritter Park 345.0 kV	Reconductor line	\$4,633,000	28% (\$1,288,000)
	Total allocated to project		\$107,373,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.